Saikat khan

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https://github.com/Saikat-dot678

https://portfolio-blond-one-75.vercel.app/

Skills

Programming & AI: — Embedded C, C++, Python, Javascript,OpenCV,DSA,OOPs.

Embedded Systems: — Raspberry Pi boards, Espressif boards.

Communication & IoT: — I2C/SPI/UART Protocols, MQTT protocol, TCP/UDP basics.

Development Tools: — Arduino IDE, Thony IDE, ESP IDF, VS Code,Docker,Anaconda.

Database: — MySQL, PostgreSQL,MongoDB

Education

National Institute of Technology Durgapur 08/2024 – present
Bachelor of Technology in Computer science and Engineering Durgapur
CGPA :- 9.2/10

Bishnupur High School 2024
Higher Secondary Education Bishnupur
Percentage :- 92.4%

Projects

Sohojia foundation Monitoring System(full stack & AI)

- We built this system to eliminate unreliable manual attendance processes and ensure **accurate, tamper-proof volunteer monitoring** for rural education programs. Automated geofencing, biometric verification, and RBAC controls give the NGO **real-time visibility, accountability, and trusted data** needed for effective program management.

- *Node.js, Python, InsightFace, MySQL, MongoDB, Docker* **Architected a hybrid microservice system** to automate volunteer monitoring for a rural education NGO, replacing manual oversight with a secure web-based platform. **Implemented Multi-Factor Attendance:** Engineered a verification pipeline using **GPS Geofencing** (100m radius) and **Face Verification** via a Python microservice (InsightFace bufalo__l model). **Developed Secure RBAC:** Built a hierarchical Role-Based Access Control system using **HttpOnlyJWT cookies** and bcrypt hashing to manage , Director, and Volunteer permissions. **Cloud & Automation:** Integrated Google Cloud Storage for biometric data and established **Automated Cron Jobs** for real- time lateness tracking and absence reporting.

IoT-Based Smart Attendance System (Embedded Systems & IoT Applications)

- We built this device to solve the problem of **unreliable and easily manipulated manual attendance** in schools, especially in areas with poor or no internet connectivity. By using offline-capable fingerprint authentication, the system ensures **accurate student verification, consistent record-keeping, and dependable operation** in any environment.

- Designed and prototyped a smart attendance device using fingerprint authentication for secure and accurate student identification.Engineered the system to fun ction offline, storing records locally without internet dependency, ensuring reliability in varied environments.

LoRa Mesh Communication System (Embedded Systems, Networking)

- We built this decentralized LoRa transport layer to overcome **high packet loss, long-range interference, and unreliable communication** in rural and off-grid environments. Existing LoRaWAN stacks couldn't provide the reliability needed at long distances (especially at SF12), so we engineered a custom dual-radio architecture and advanced ARQ mechanisms to ensure **robust data delivery, continuous reception, and collision-resilient networking** across expanding multi-node systems.

- *ESP32-S3, Dual SX1262, Python, SPI, Selective Repeat ARQ* **Developed a decentralized Long Range (LoRa) protocol** a **Reliable Transport Layer** to ensure data integrity over long-distance, high-interference . **Designed Dual -Radio Full-Duplex Emulation:** Configured a dual SX1262 hardware architecture where one radio is dedicated to continuous RX and the other to TX/CAD to eliminate communication "blind spots" **Implemented Advanced Reliability:** Built a custom networking stack including **Selective Repeat ARQ**, packet fragmentation, and CRC16-CCITT checksums to mitigate packet loss at Spreading Factor 12 (SF12). **Engineered Collision Avoidance:** Integrated a **Listen Before Talk (LBT)** algorithm with random backof and Channel Activity Detection (CAD) to manage medium access in congested mesh environments. . Currently working on expanding the network to **multi-node system.**

Position and Responsibility

CSE Students' Society - NIT Durgapur 09/2025 – Present
Junior member

Robotica (IIC) - NIT Durgapur 05/2025 – present
Junior member

Achivements

Winner in **Burn Brain 2025** hackathon organized by CSS NIT Durgapur 04/2025

Winner in Aarohan's **On Spot Hackathon 2025** organized by CCA Robocell and GLUG NIT Durgapur 03/2025